SHELLVIVE

Prototype 1

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Game Overview

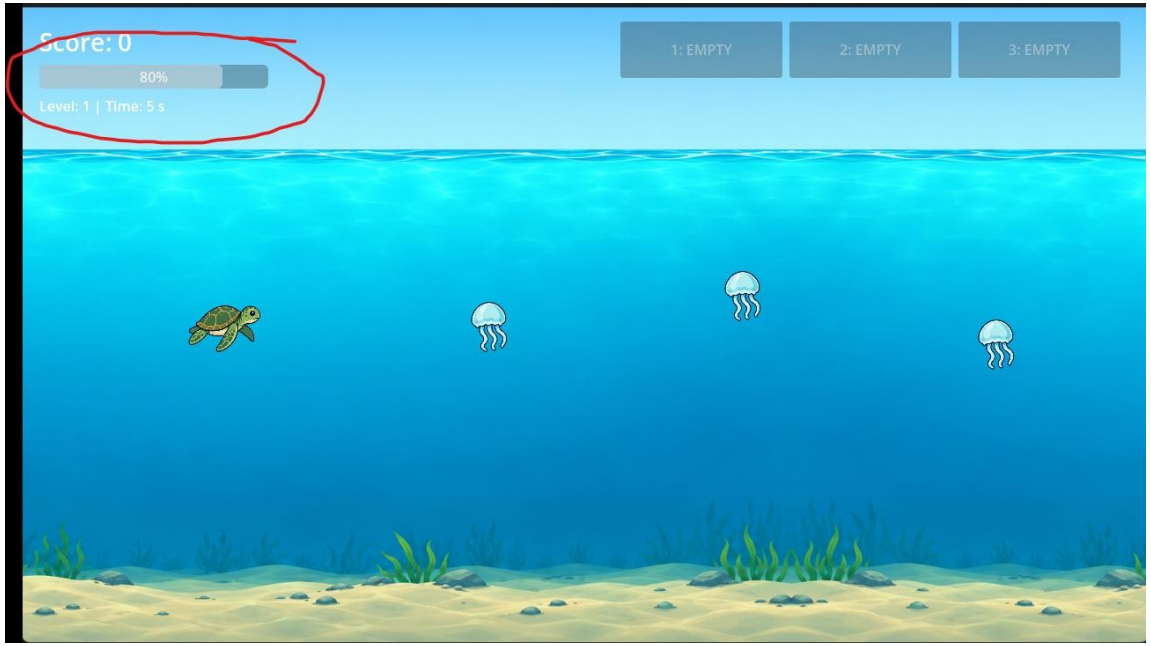
Shellvive is a single-player 2D survival game about ocean plastic pollution. Players control a sea turtle, eat jellyfish to maintain its survival bar, avoid plastic hazards and collect assistance cards.

As pollution increases, survival becomes more difficult. The game aims to highlight the threats plastic waste poses to marine life.

Game Objective

Survive for as long as possible and collect points.

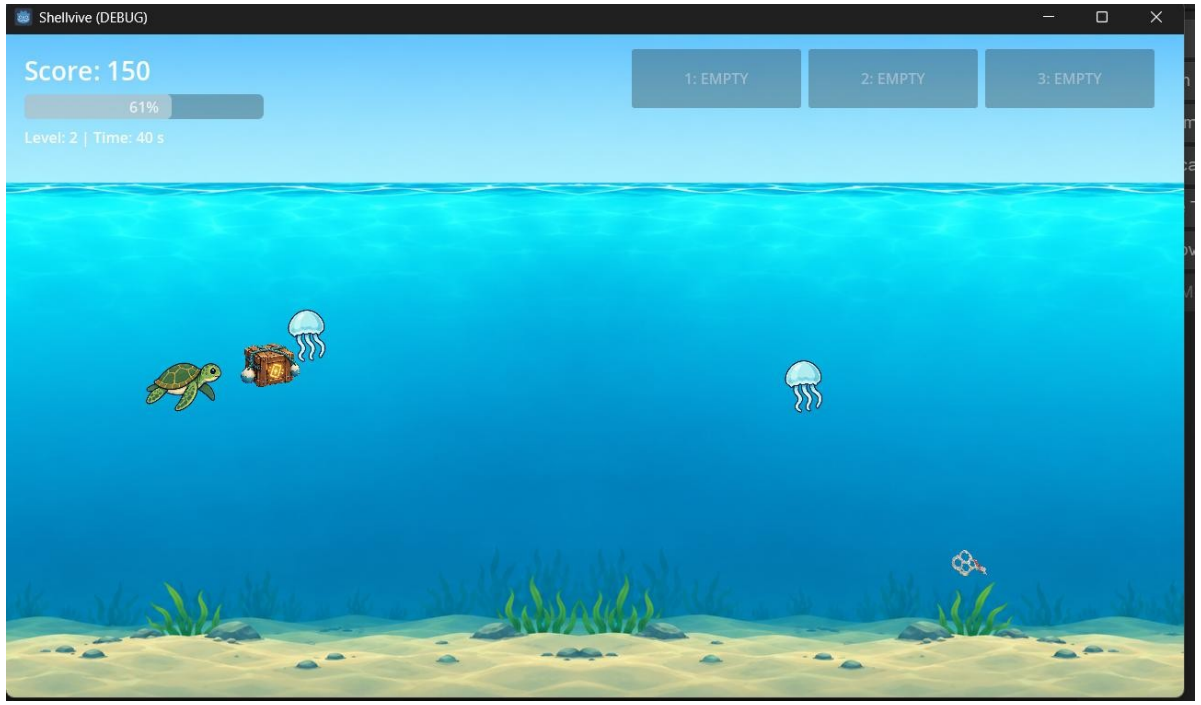
- Eat jellyfish to restore the survival bar.
- Avoid plastic that slows the turtle or reduces survival.
- Collect and use assistance cards when needed.



The game ends when the survival bar reaches zero.

Implemented Features

- Turtle movement in four directions.
- Ordinary and large jellyfish with different rewards.
- Choking and survival-reducing plastic.
- Floating boxes that reveal assistance cards.
- An inventory of up to three cards.
- Survival, score and difficulty displays.
- Game-over and restart controls.



Prototype 1 Findings

The core gameplay loop is implemented: move, collect food, avoid plastic and use assistance.

Plastic introduces different threats, while increasing pollution makes survival harder. Cards give players limited help and a choice about when to use it.

This prototype provides a basis for playtesting. Difficulty, spawning, recovery amounts and slowdown strength still need adjustment.

Social Good Integration

Shellvive aims to raise awareness of ocean plastic pollution through its gameplay. Plastic can slow the turtle or reduce its survival bar, making food harder to reach and survival more difficult.

Rescue and cleanup cards provide temporary help, but new plastic continues to appear. This shows why preventing pollution matters alongside cleanup efforts.

The game supports SDG 14: Life Below Water and connects to SDG 12: Responsible Consumption and Production.

Plans for Prototype II

- Adjust difficulty and spawning based on playtest feedback.
- Improve food collection feedback and card controls.
- Create our own artwork and simple three-frame animations.
- Add boat animations to show assistance and pollution events.
- Develop additional card effects.

Boat animations and the planned death sequence are not yet implemented.

Prototype 1 AI Usage

ChatGPT generated the prototype code and placeholder artwork.

We imported the assets, connected scenes and scripts, configured movement inputs, adjusted collision shapes and sprite sizes, and tested the game in Godot.

The current visuals are placeholders. We plan to replace them with our own artwork.

Team Contributions

Yangli Deng - approximately 50%

Turtle and jellyfish scenes, movement inputs, collision shapes, background and survival/score displays. Integration and testing of movement, eating and restart behaviour.

Sarah Ali - approximately 50%

Plastic and box scenes, card interface, spawning and difficulty settings. Integration and testing of plastic effects and assistance cards.

Shared work

Game design, AI prompts and asset selection, scene connections, playtesting and planning improvements.